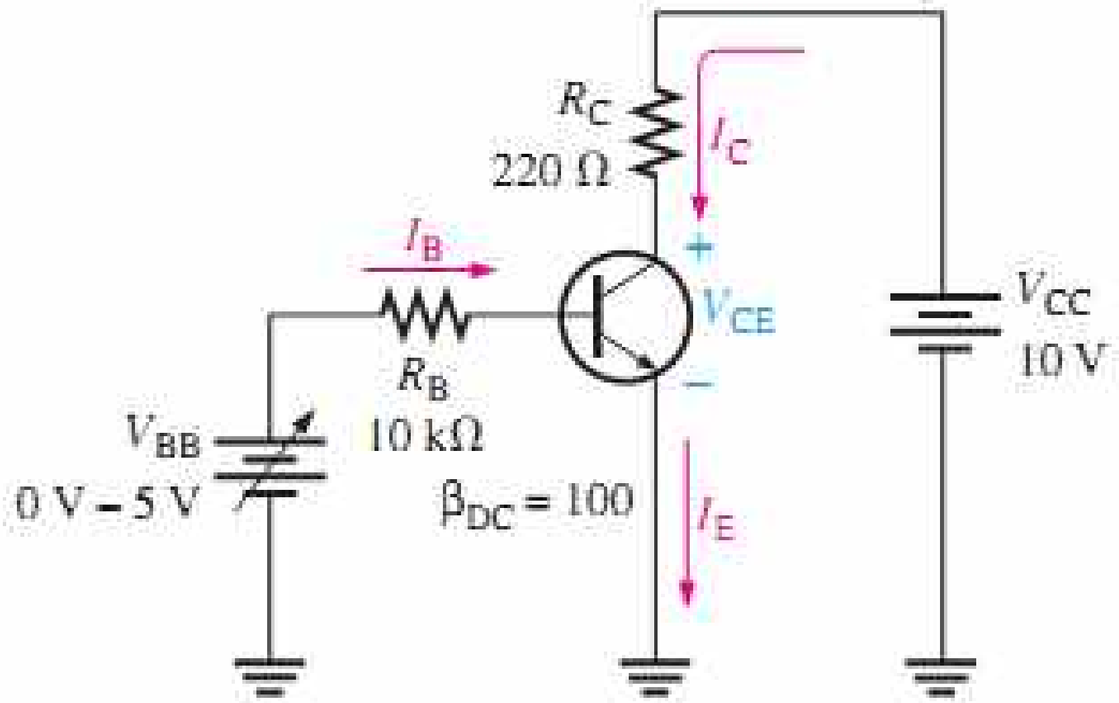
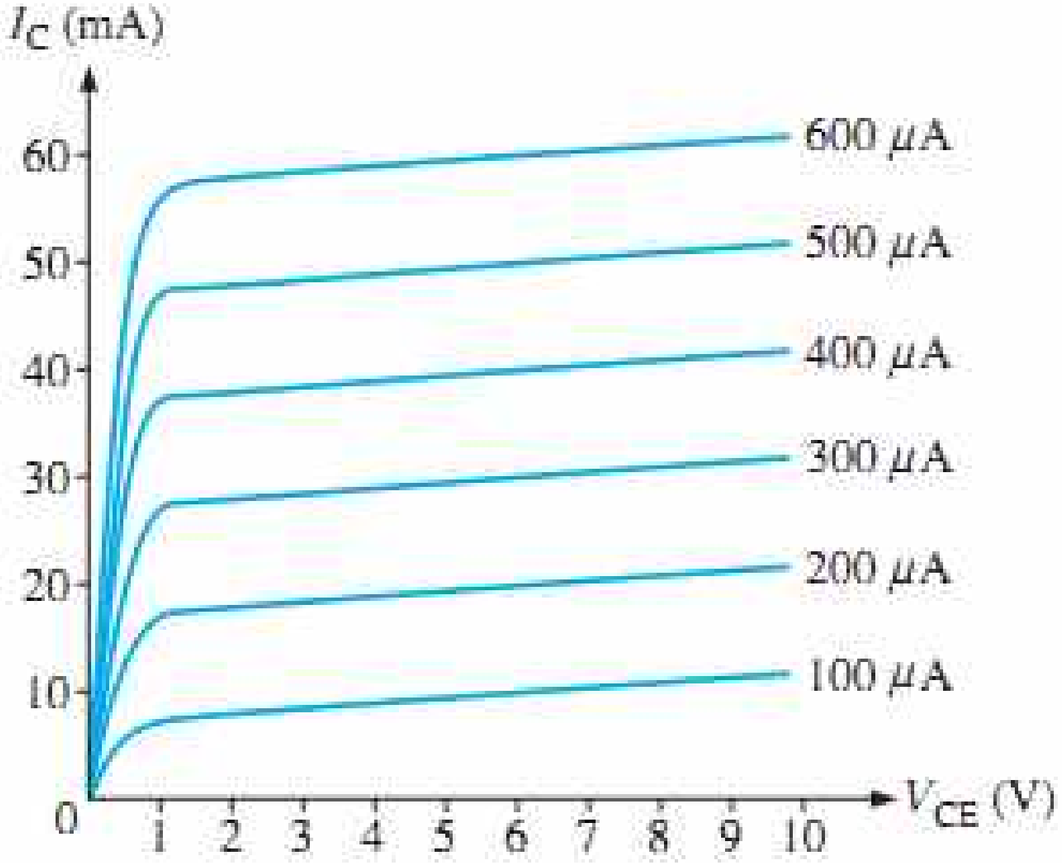
EGCI 232 Engineering Electronics

- Common Emitter Amplifier

Biasing NPN Transistor

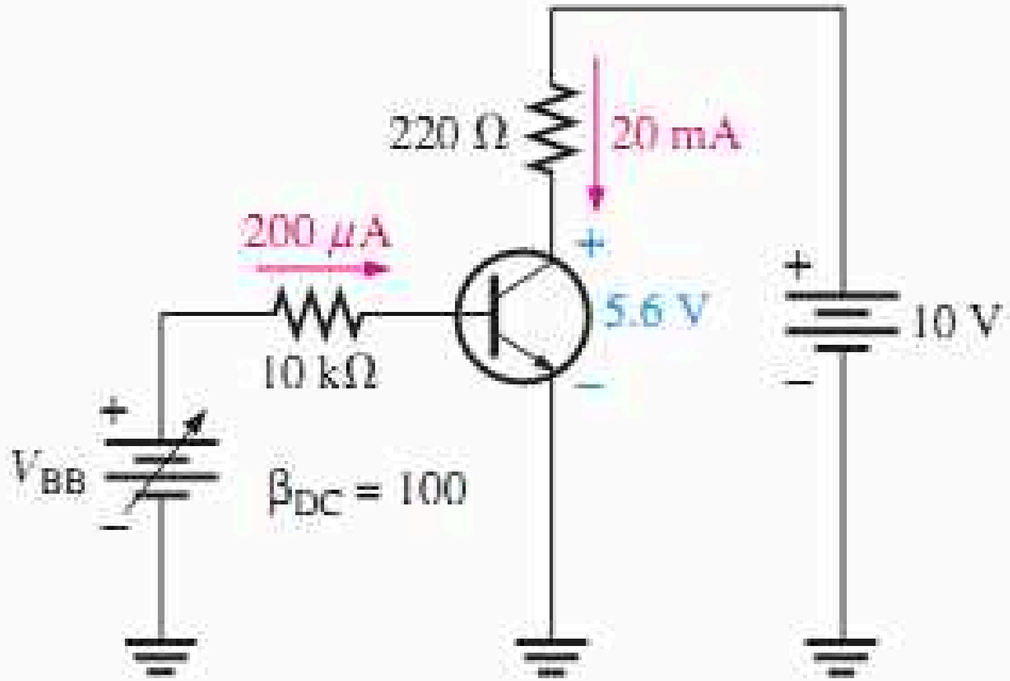


(a) DC biased circuit

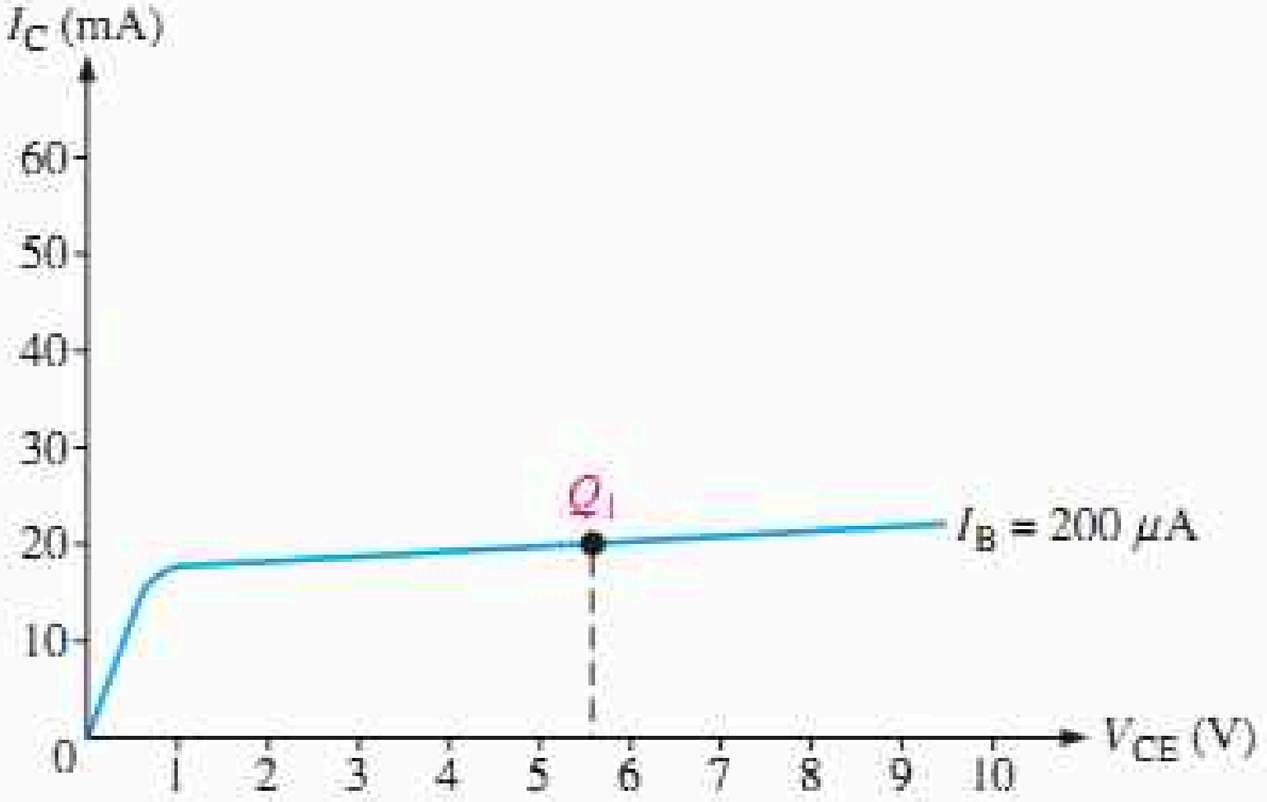


(b) Collector characteristic curves

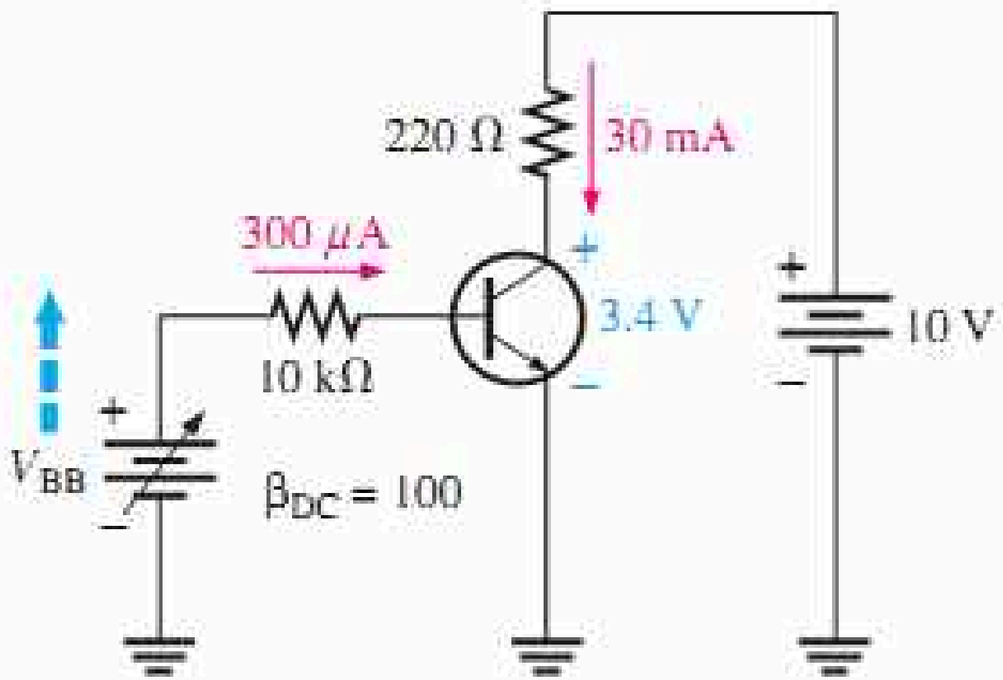
Biasing NPN Transistor



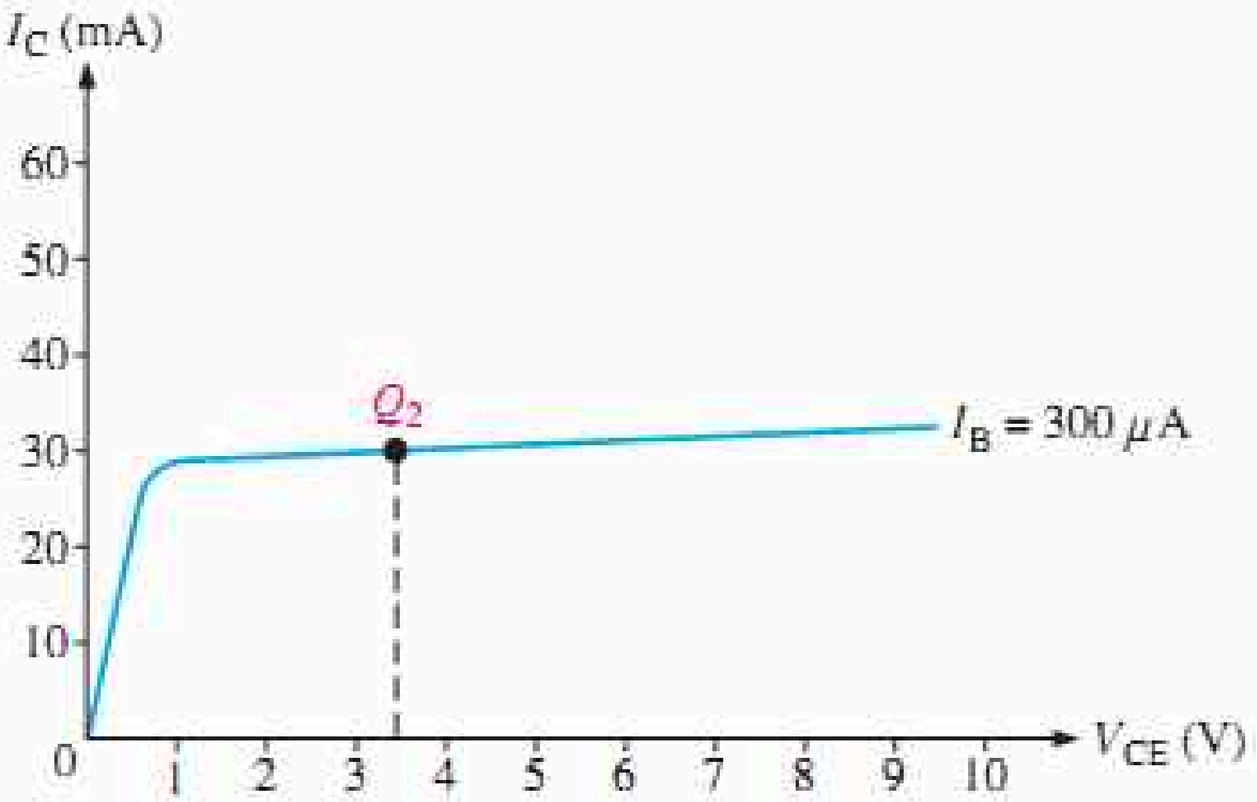
(a) $I_B = 200\ \mu A$



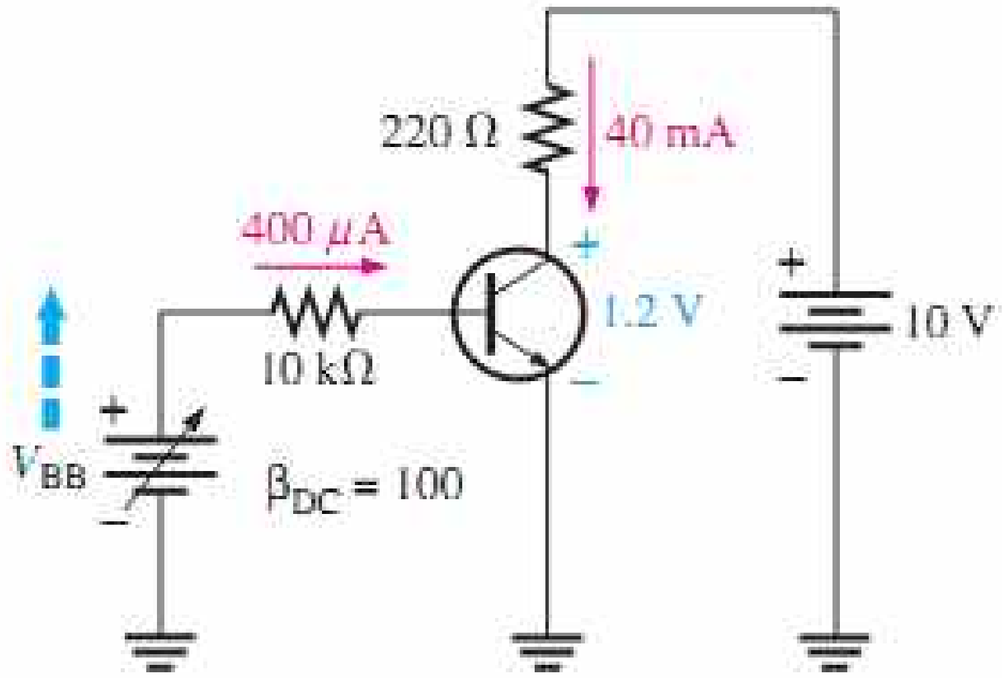
Biasing NPN Transistor



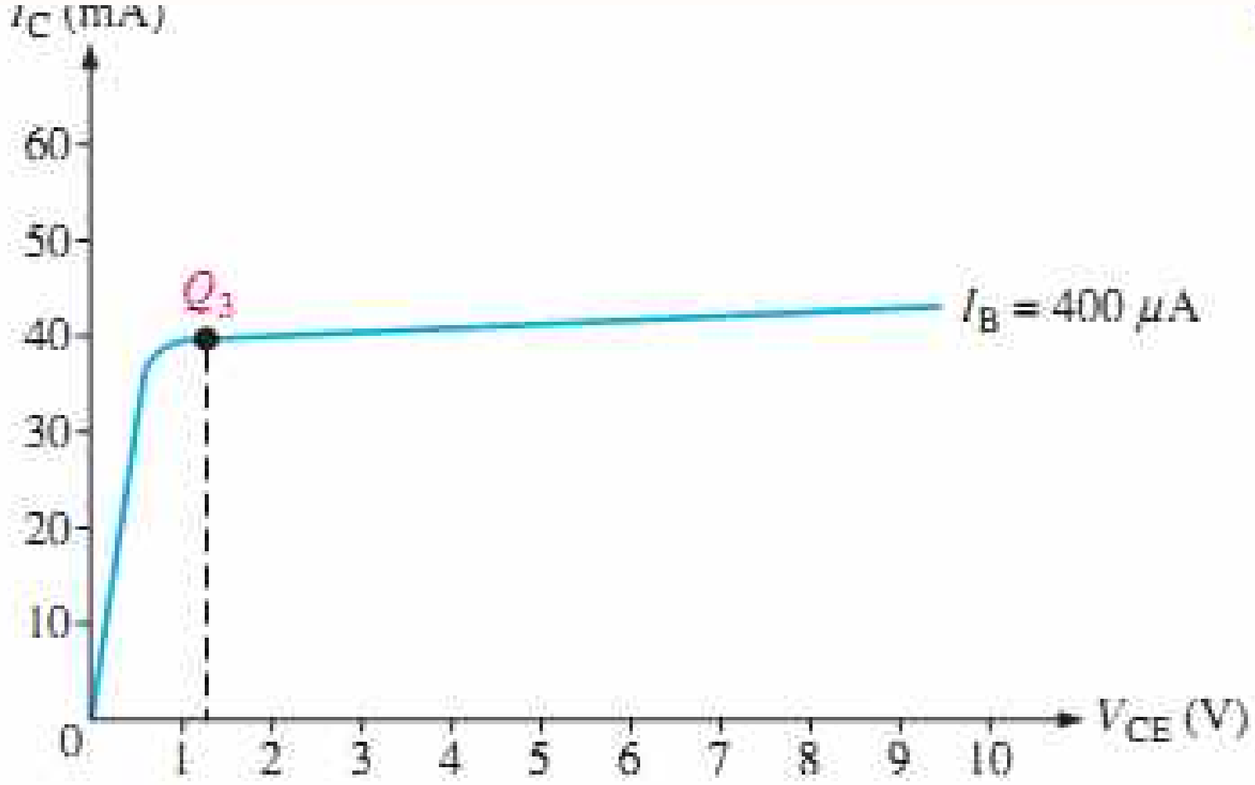
(b) Increase I_B to 300 μA by increasing V_{BB}



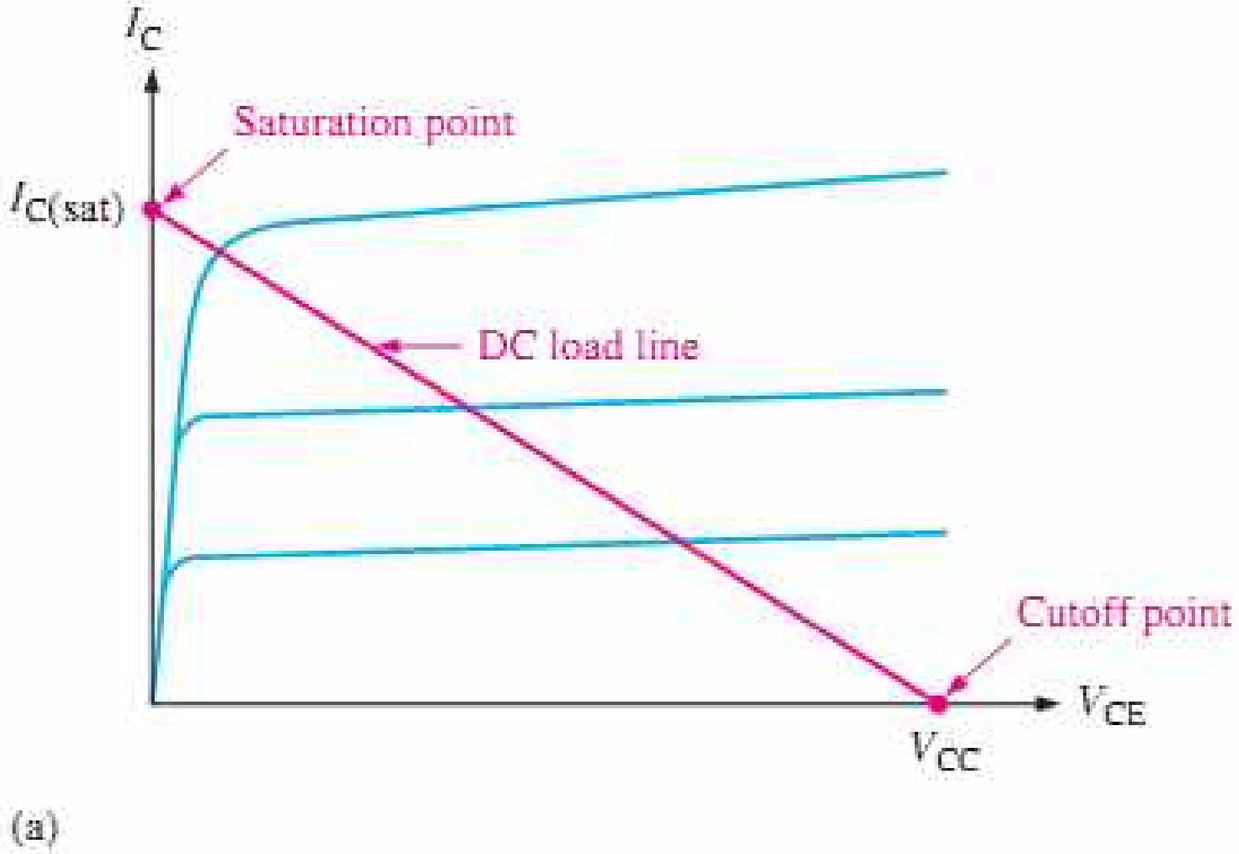
Biasing NPN Transistor



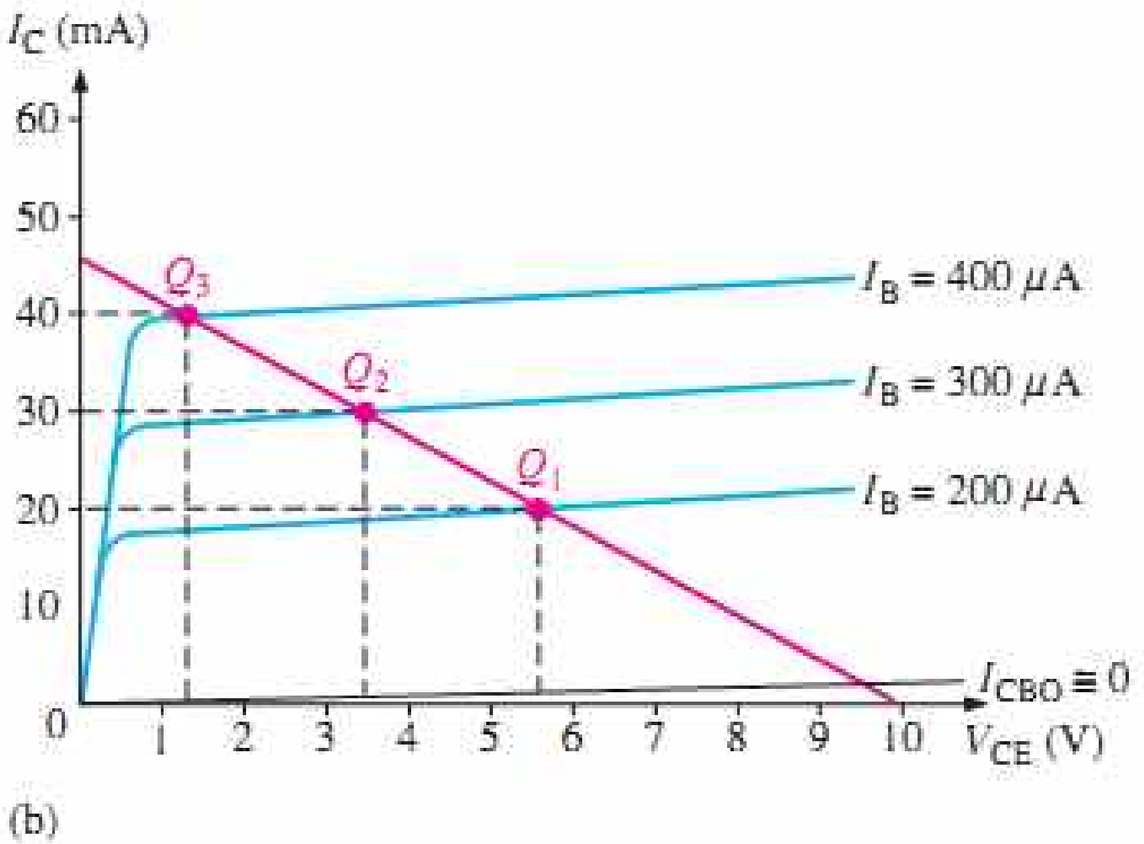
(c) Increase I_B to 400 μA by increasing V_{BB}



Load Line

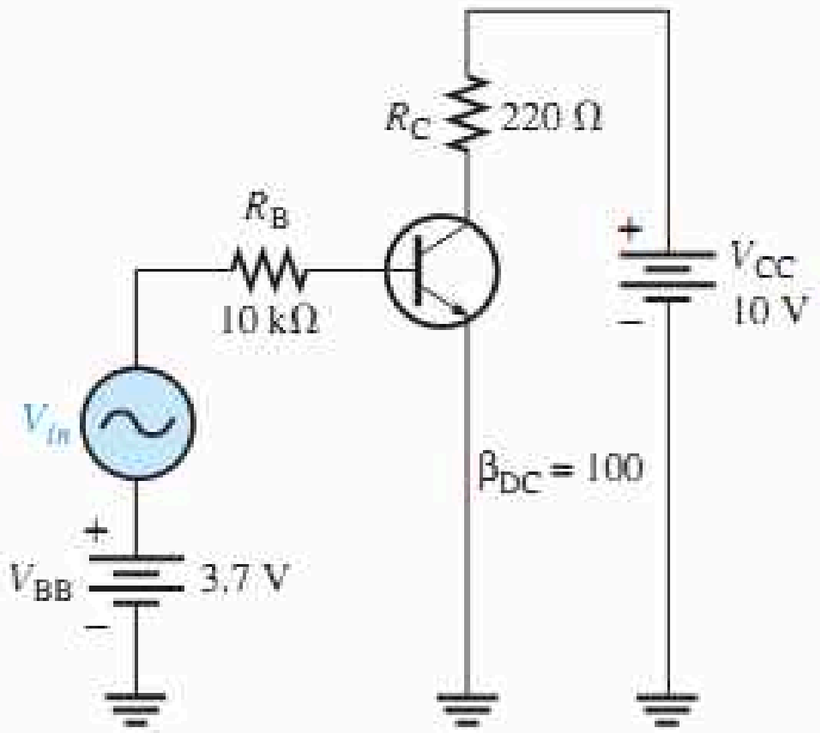

I_C
Saturation point
$I_{C(sat)}$
DC load line
Cutoff point
V_{CE}
V_{CC}

(a)


I_C (mA)
60
50
40
30
20
10
0
Q_3
Q_2
Q_1
$I_B = 400\ \mu A$
$I_B = 300\ \mu A$
$I_B = 200\ \mu A$
$I_{CBO} \cong 0$
1 2 3 4 5 6 7 8 9 10
V_{CE} (V)

(b)

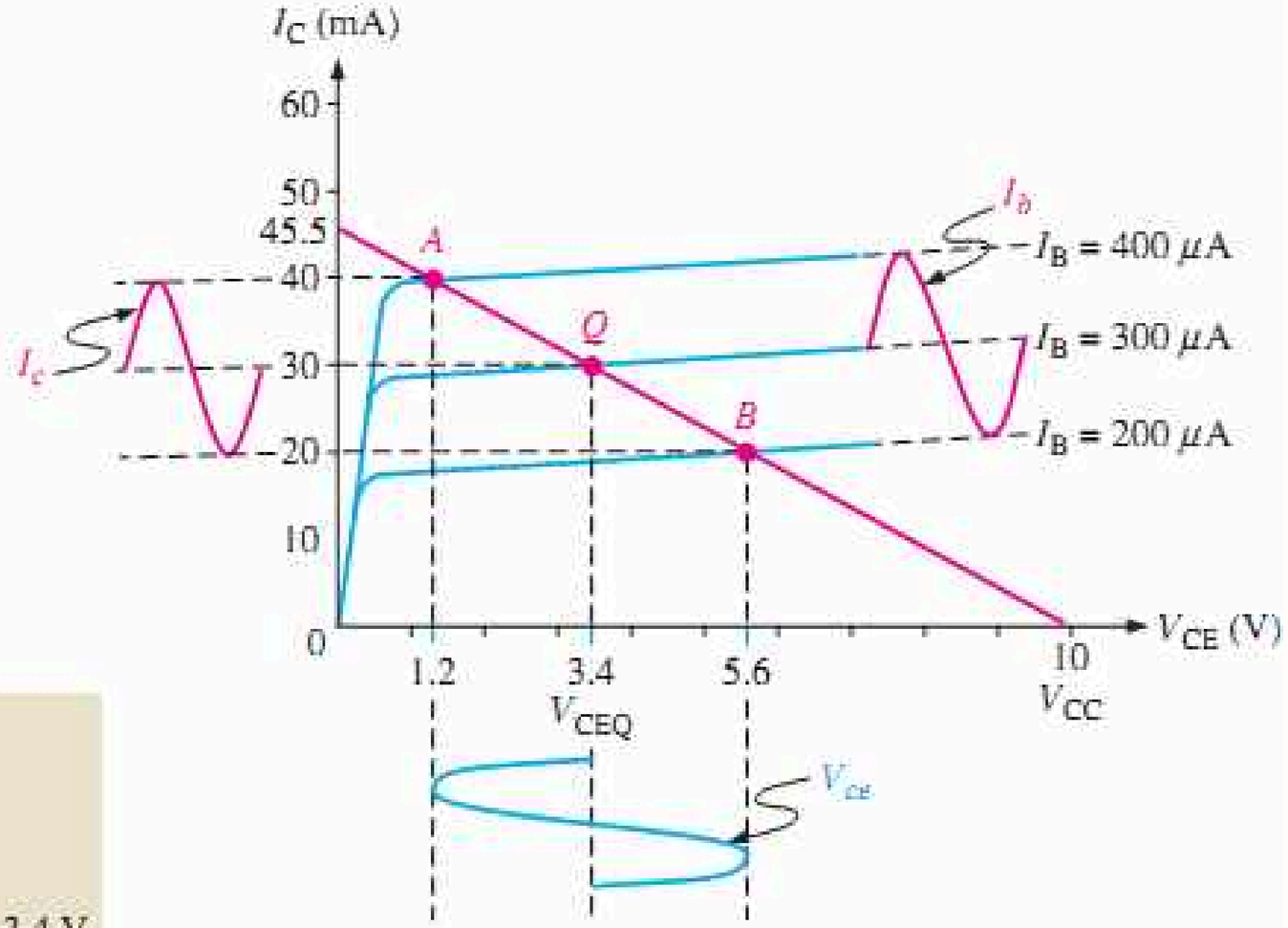
Q-point



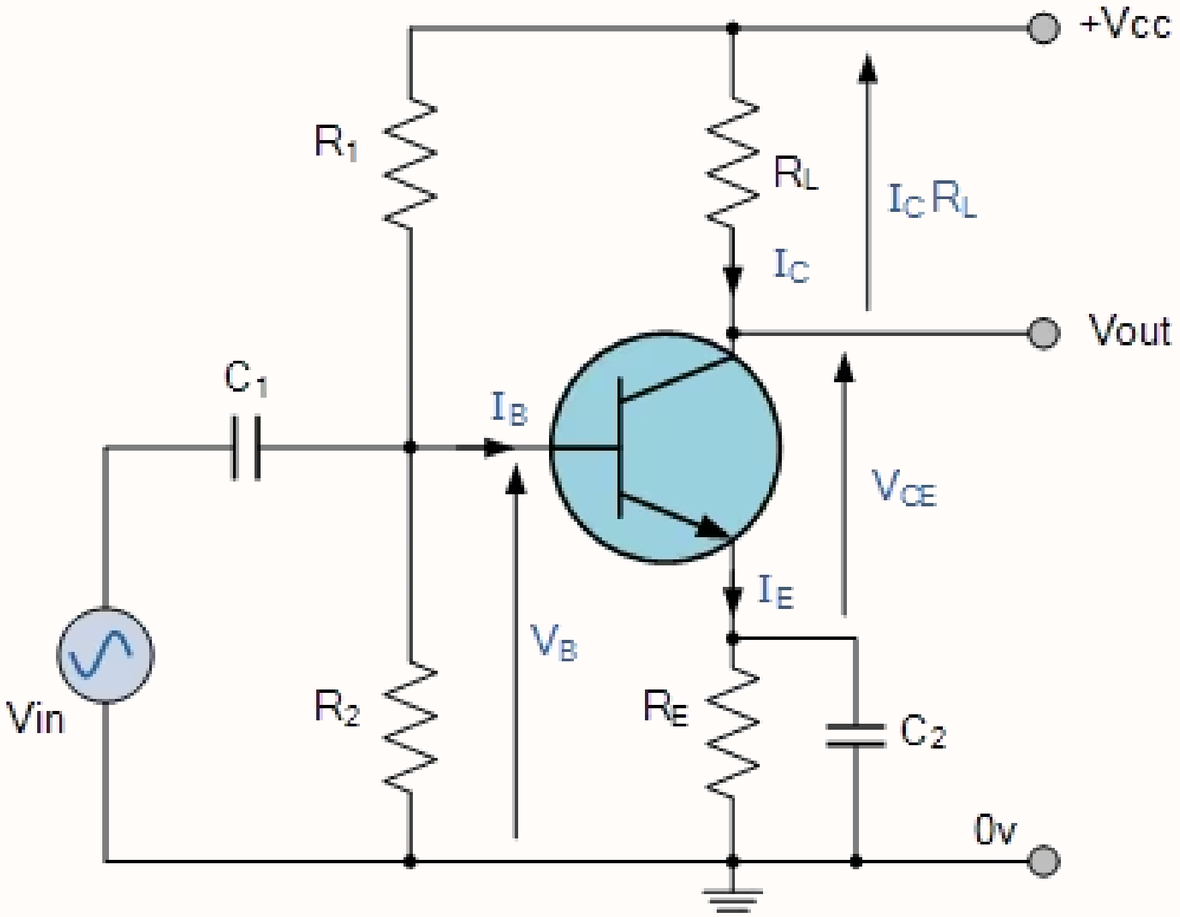
$$I_{BQ} = \frac{V_{BB} - 0.7\text{ V}}{R_B} = \frac{3.7\text{ V} - 0.7\text{ V}}{10\text{ k}\Omega} = 300\ \mu\text{A}$$

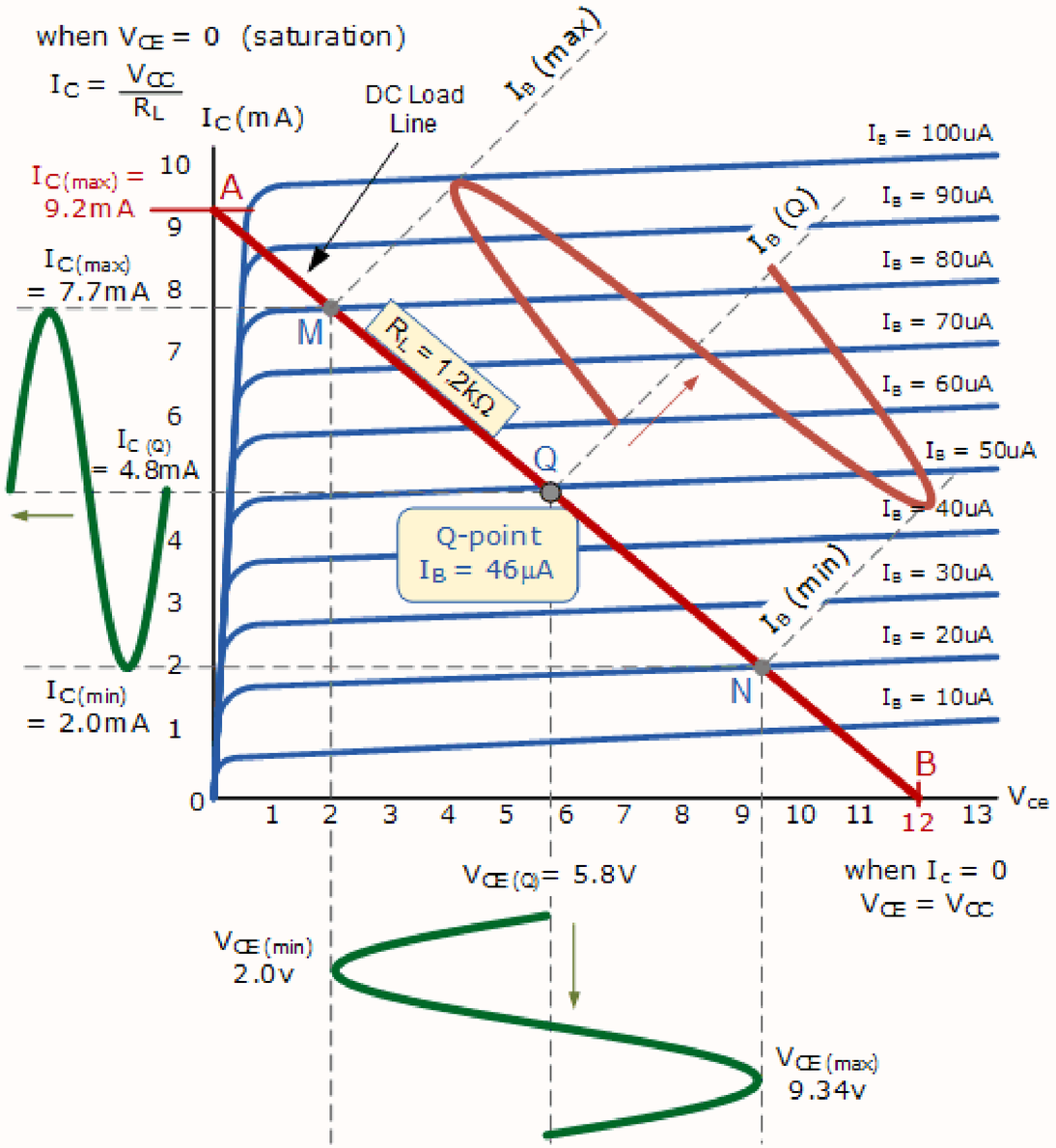
$$I_{CQ} = \beta_{DC} I_{BQ} = (100)(300\ \mu\text{A}) = 30\text{ mA}$$

$$V_{CEQ} = V_{CC} - I_{CQ} R_C = 10\text{ V} - (30\text{ mA})(220\ \Omega) = 3.4\text{ V}$$

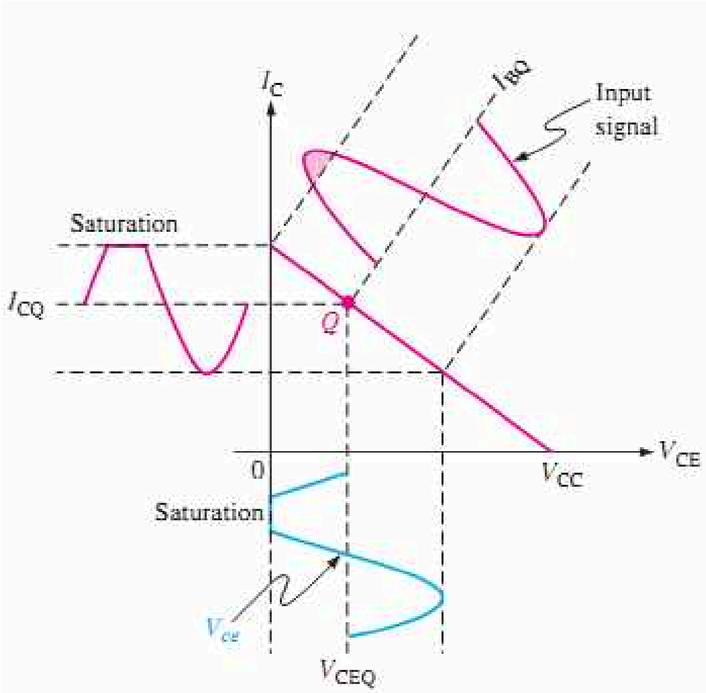


Amplifier

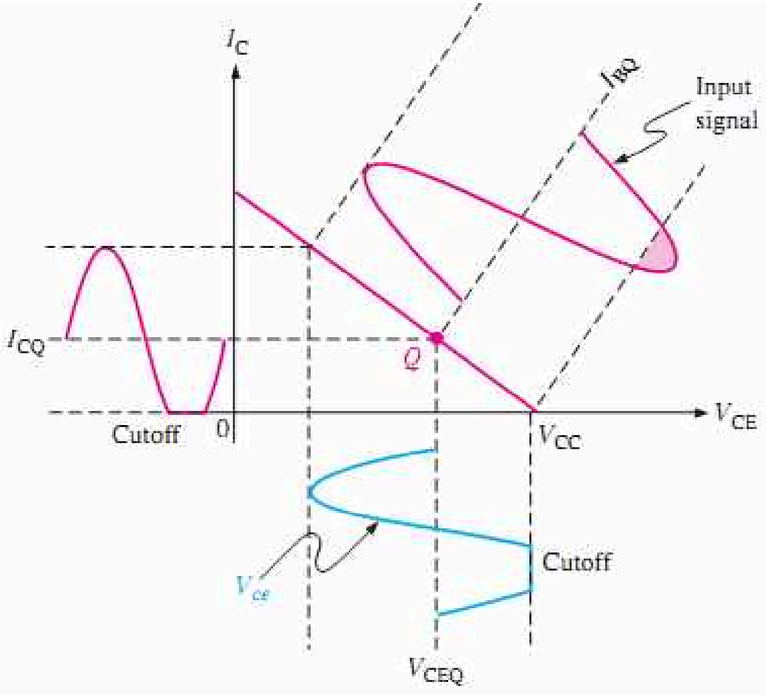

+Vcc
R1
RL
IC RL
IC
Vout
C1
IB
VCE
IE
VB
Vin
R2
RE
C2
0v


when VCE = 0 (saturation)
IC = Vcc / RL
IC(mA)
DC Load Line
IB (max)
IB = 100uA
IB = 90uA
IB = 80uA
IB = 70uA
IB = 60uA
IB = 50uA
IB = 40uA
IB = 30uA
IB = 20uA
IB = 10uA
IB (Q)
IB (min)
IC(max) = 9.2mA
IC(max) = 7.7mA
IC (Q) = 4.8mA
IC(min) = 2.0mA
A
M
Q
N
B
RL = 1.2kΩ
Q-point
IB = 46µA
Vce
VCE(Q) = 5.8V
when Ic = 0
VCE = Vcc
VCE(min) 2.0v
VCE(max) 9.34v

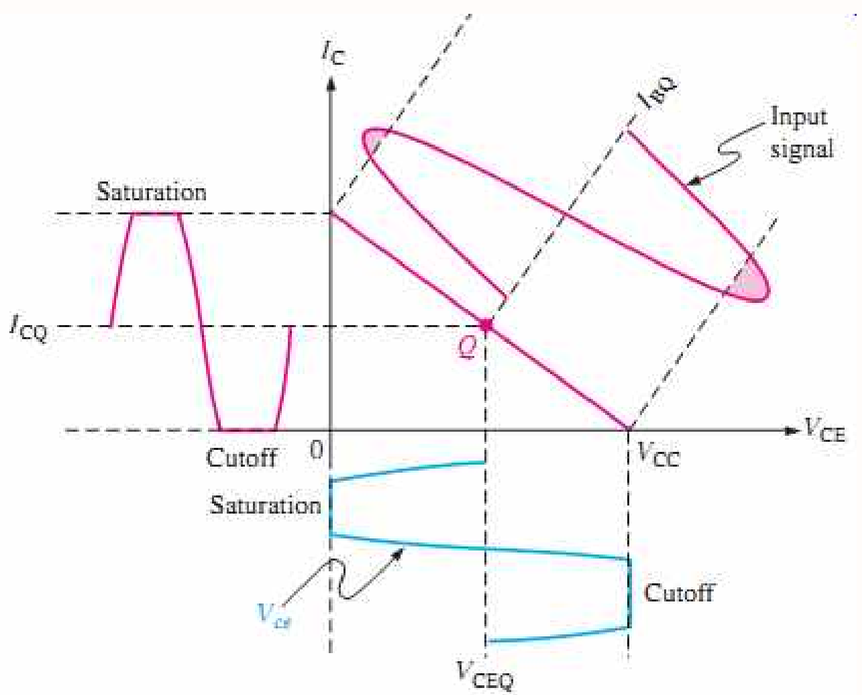
Amplifier



(a) Transistor is driven into saturation because the Q-point is too close to saturation for the given input signal.

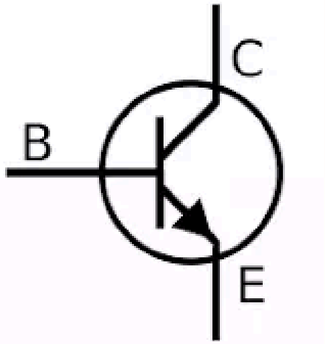


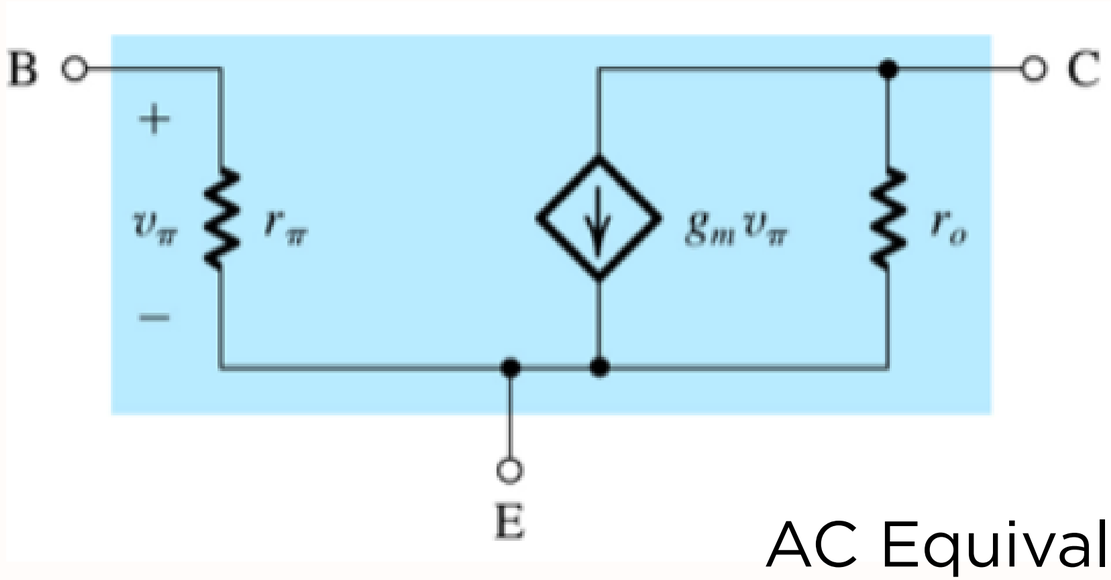
(b) Transistor is driven into cutoff because the Q-point is too close to cutoff for the given input signal.



(c) Transistor is driven into both saturation and cutoff because the input signal is too large.

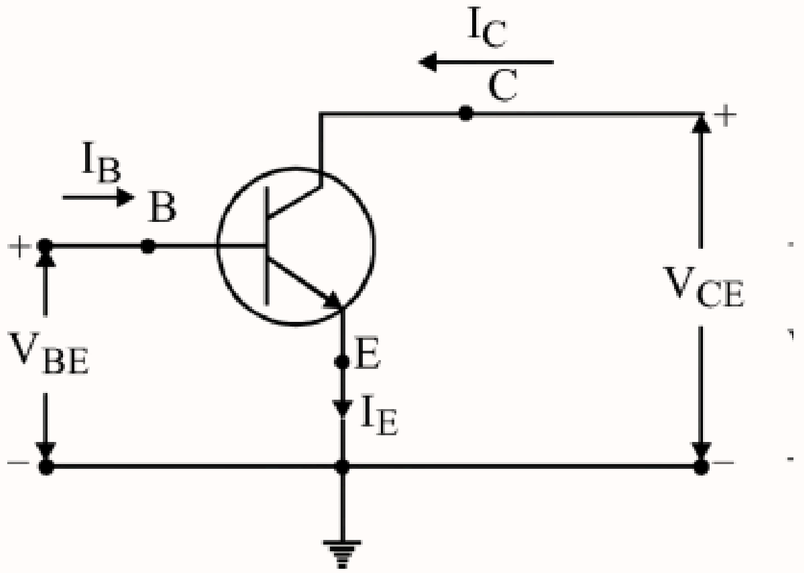
Analyzing amplifier circuit





AC Equivalent Circuit

Two Port Network

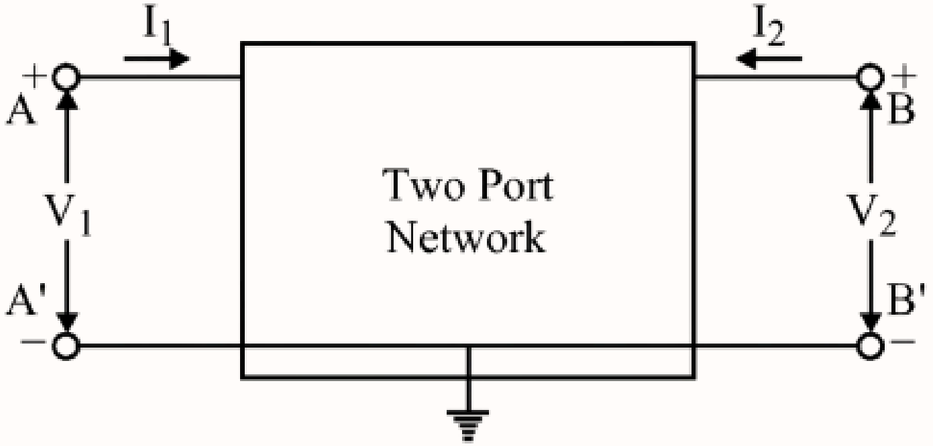


I_B is the input current to the transistor.

V_{BE} is the input voltage to the transistor.

I_C is the output current of the transistor.

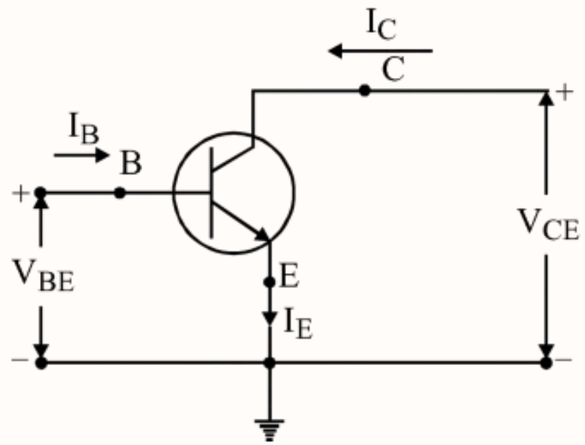
V_{CE} is the output voltage of the transistor.

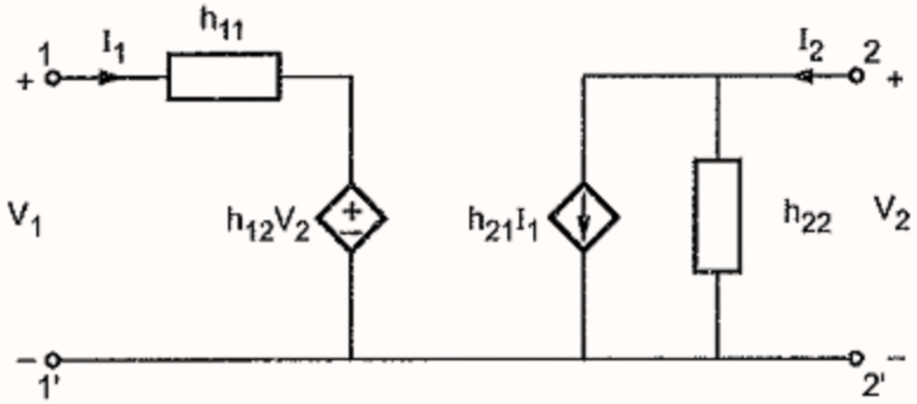


$$V_1 = f_1(I_1, V_2)$$
$$I_2 = f_2(I_1, V_2)$$

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

Two Port Network





The individual h-parameters are defined as follows,

$$h_{11} = \left.\frac{V_1}{I_1}\right|_{V_2 = 0}$$

This is called short circuit input impedance.

$$h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2 = 0}$$

This is called forward short circuit current gain.

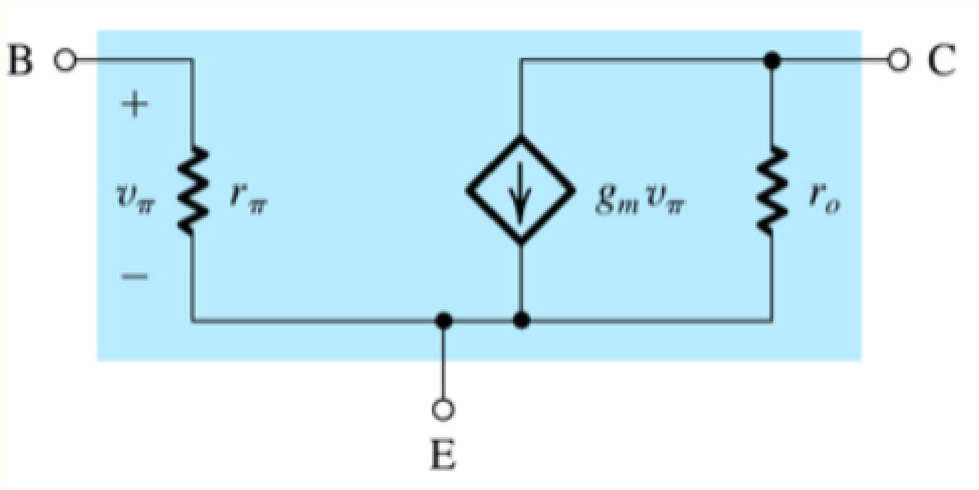
$$h_{12} = \left.\frac{V_1}{V_2}\right|_{I_1 = 0}$$

This is called reverse open circuit voltage gain.

$$h_{22} = \left.\frac{I_2}{V_2}\right|_{I_1 = 0}$$

This is called open circuit output admittance.

Hybrid-pi (ac) model of NPN transistor



Parameter	Description	Condition
$h_{11} = V_1 / I_1$	Input impedance	$V_2 = 0$ (output short-circuited)
$h_{12} = V_1 / V_2$	Reverse voltage gain	$I_1 = 0$ (input open-circuited)
$h_{21} = I_2 / I_1$	Forward current gain	$V_2 = 0$ (output short-circuited)
$h_{22} = I_2 / V_2$	Output admittance	$I_1 = 0$ (input open-circuited)

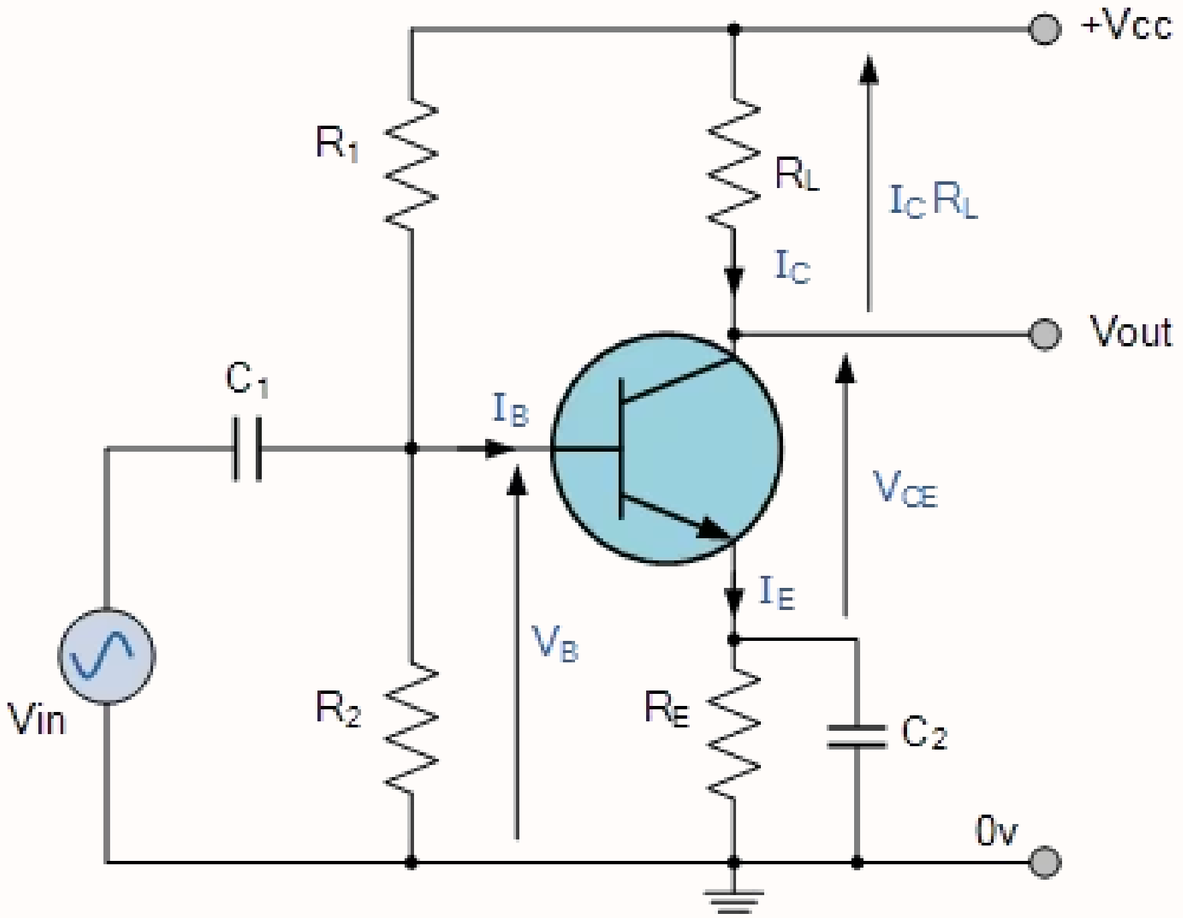
AC Analysis

1) Input impedance (Zi).

2) Output impedance (Zo).

3) Voltage gain (Av). &

4) Current gain (Ai).



AC Analysis

Example

1. Design the common emitter amplifier circuit shown in Figure 1
   a. Determine R_C, R_E, R_1, and R_2 such that the transistor is DC biased to I_C = 1mA and V_{CE} = 6V
   b. Use the method of $I_{R1} = 10I_B$, and $I_{R2} = 9I_B$ to determine R1 and R2).
   c. Using small signal equivalents of this amplifier circuit, calculate the voltage gains $A_{vi} = (V_o/V_{in})$ and $A_{vs} = (V_o/V_s)$, and the input resistance R_i, and the output resistance R_o.

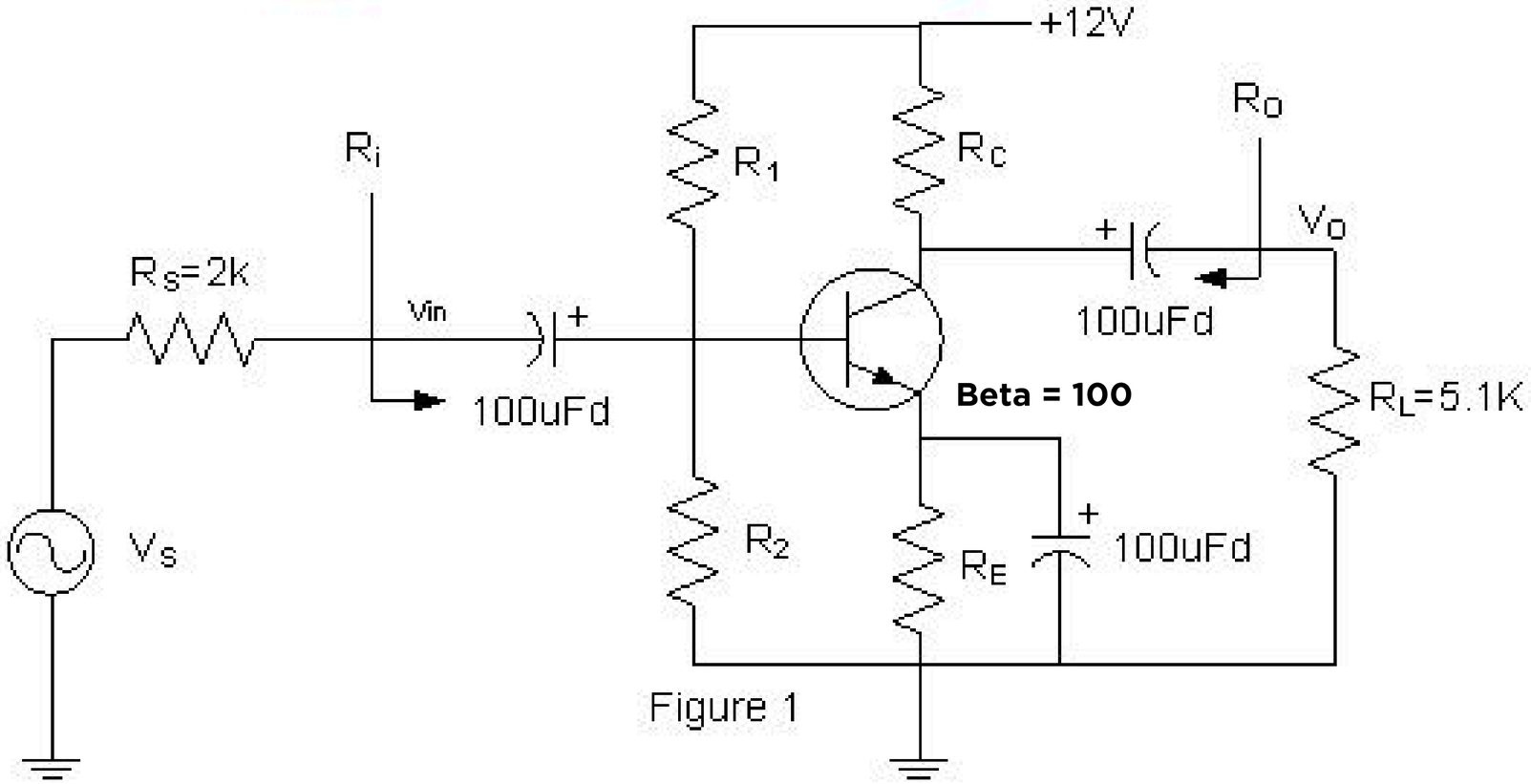


Figure 1

DC Analysis

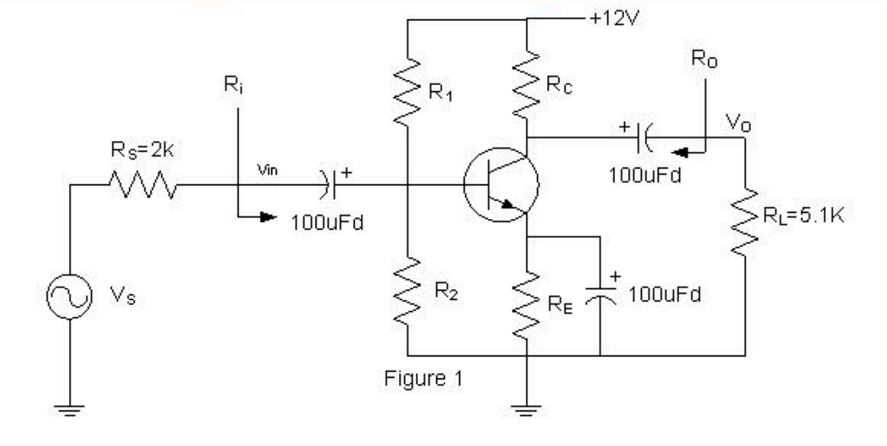


Figure 1

AC Analysis

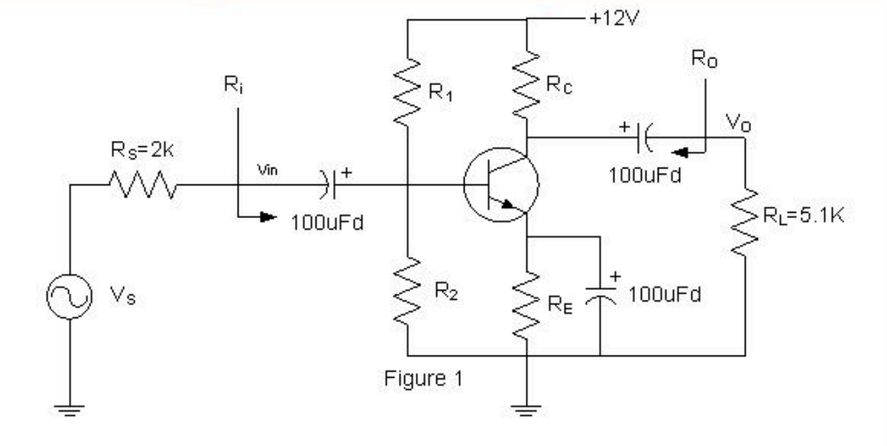
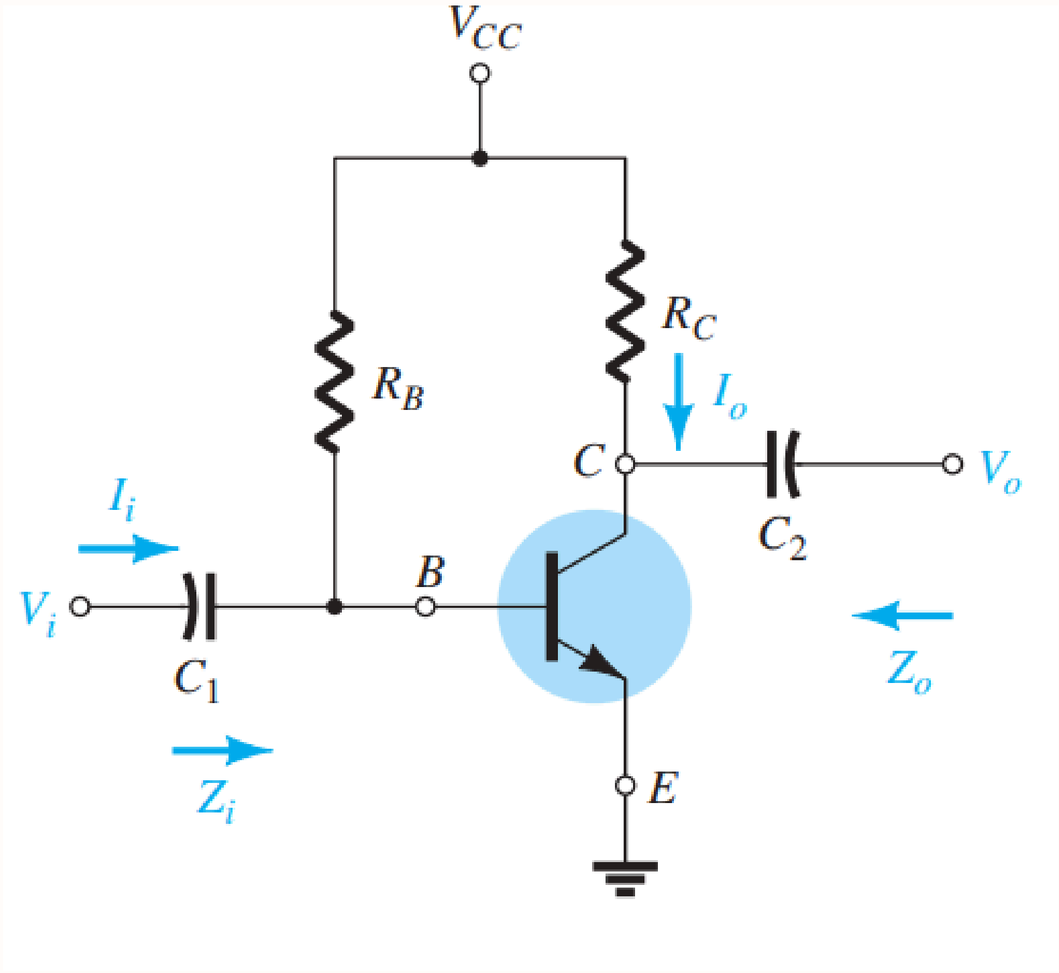


Figure 1

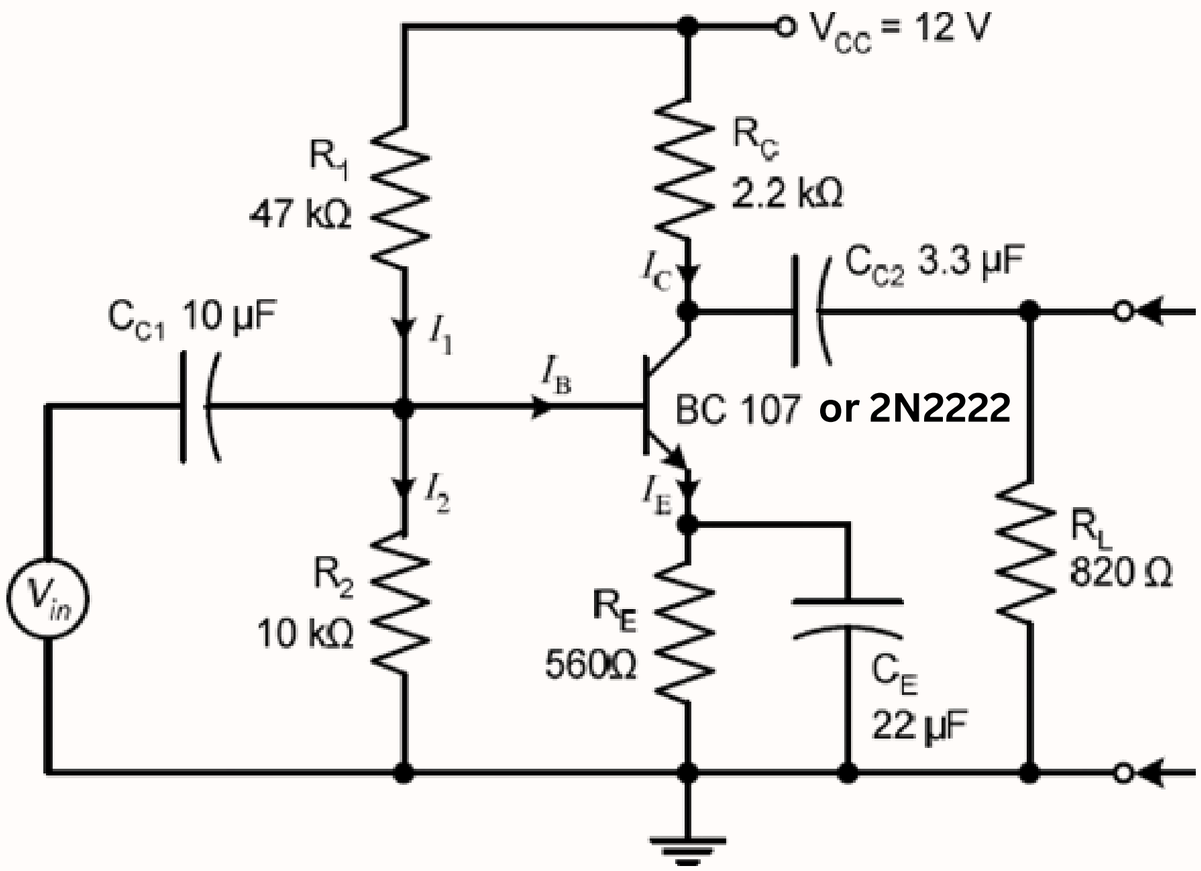
Example



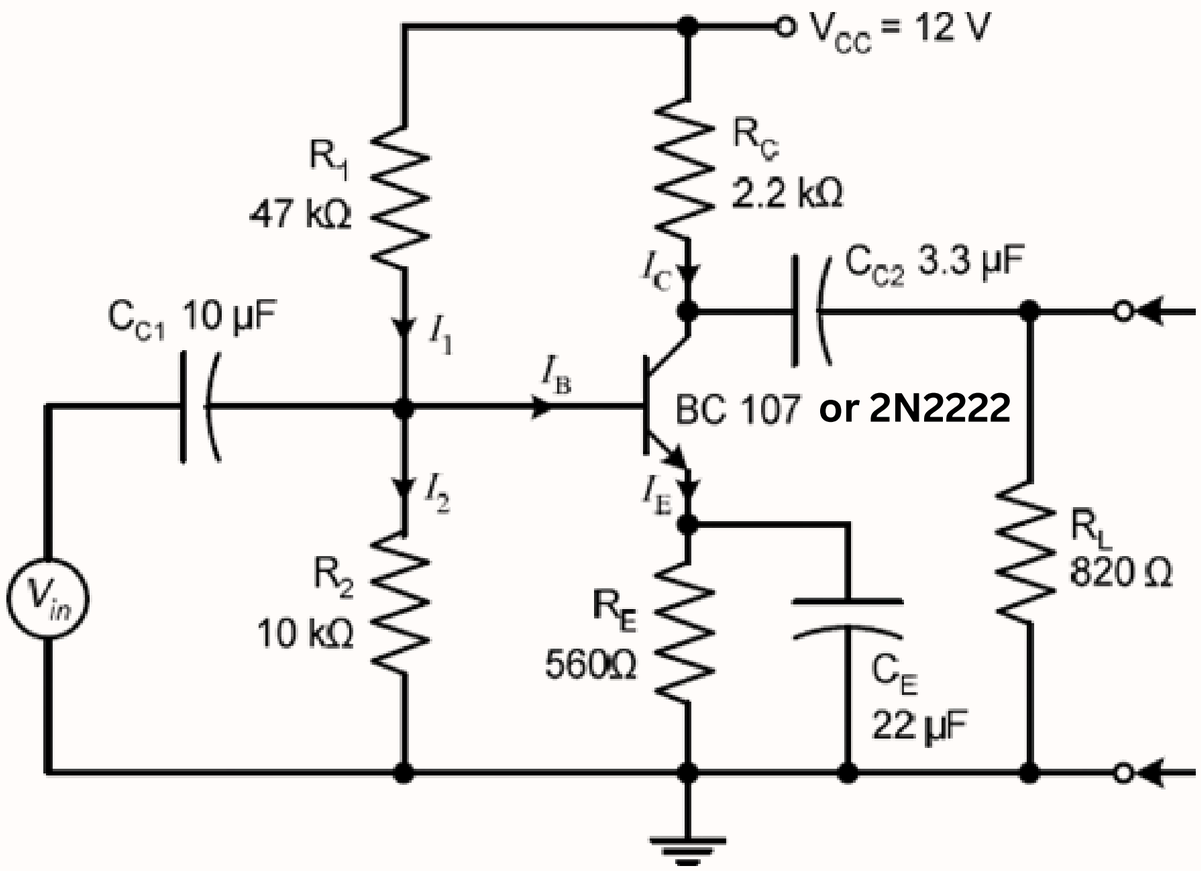
Example

Example

Experiment


V_{CC} = 12 V
R_1
47 kΩ
R_C
2.2 kΩ
I_C
C_{C2} 3.3 µF
C_{C1} 10 µF
I_1
I_B
BC 107 or 2N2222
I_2
I_E
R_L
820 Ω
V_{in}
R_2
10 kΩ
R_E
560Ω
C_E
22 µF

Experiment


V_{CC} = 12 V
R_1
47 kΩ
R_C
2.2 kΩ
I_C
C_{C2} 3.3 µF
C_{C1} 10 µF
I_1
I_B
BC 107 or 2N2222
I_2
I_E
R_L
820 Ω
V_{in}
R_2
10 kΩ
R_E
560Ω
C_E
22 µF

Experiment

Experiment